A Dual-Stream Cognitive Diagnosis Framework via Mastery-Fluency Disentanglement

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Abstract. Cognitive diagnosis models traditionally estimate student proficiency solely from binary response correctness, which conflates conceptual mastery with procedural dynamics. This paper proposes MFCD, a dual-stream cognitive diagnosis framework that disentangles mastery from fluency through multi-task learning. The cognitive stream adopts a standard CDM to produce base logits for cognitive mastery, while the behavioral stream extracts exposure intensity, empirical accuracy, and response consistency from historical interactions to capture behavioral fluency. A dynamic fusion gate is further applied to adaptively merge the two streams, and auxiliary reconstruction tasks preserve behavioral information in fluency representations via multi-task optimization. Extensive experiments on six benchmarks validate the effectiveness of MFCD across various backbone models.

Keywords: Cognitive Diagnosis · Behavioral Fluency · Fusion Gate · Multi-Task Optimization · Educational Data Mining.

1 Introduction

Cognitive diagnosis (CD) is a fundamental task in educational data mining that aims to evaluate students' latent cognitive states based on their historical learning interactions and assessment records, as shown in Figure 1. As a core driving mechanism of intelligent tutoring systems and massive open online courses, Cognitive Diagnosis Models (CDMs) provide essential quantitative support for personalized learning recommendations, adaptive testing, and intelligent curriculum design [1].

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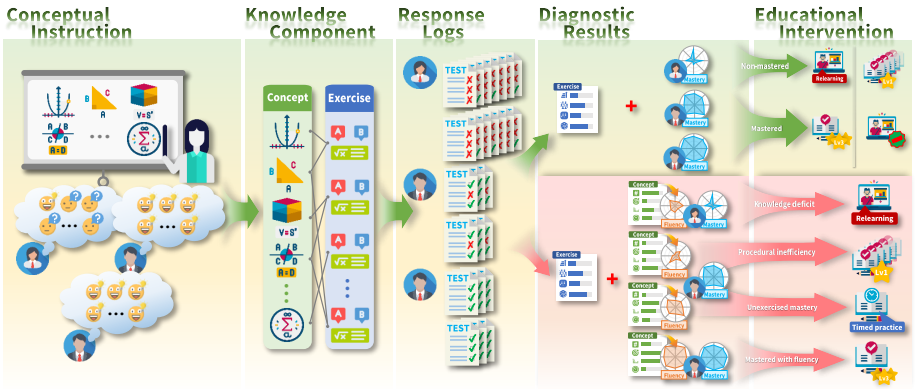


Fig. 1: A toy example of typical CD and mastery-fluency CD.

CD research has evolved from traditional psychometric models to machine learning-based approaches, and more recently to deep learning-based architectures. Psychometric-based models [2–5] provide strong theoretical interpretability through explicit statistical modeling of cognitive states, but fail to capture complex learning dynamics due to rigid parametric assumptions. Learning latent cognitive states from historical interaction logs in a data-driven manner, machine learning-based models [6–9] typically rely on manually designed features and shallow architectures, whereas deep learning-based models [10–12] also leverage neural networks to automatically capture complex, non-linear dependencies.

As illustrated in the upper branch of Figure 1, the typical CD pipeline maps response logs to a unidimensional mastery profile, yielding only binary decisions such as mastered or non-mastered. This diagnosis paradigm relies almost exclusively on binary correctness logs, reducing cognitive states to a single mastery dimension while overlooking the rich behavioral dynamics inherent in the learning process. Pedagogy and cognitive psychology [13] distinguish declarative knowledge (i.e., knowing a concept) from procedural knowledge (i.e., fluently solving a problem) as qualitatively distinct cognitive dimensions, and conflating a lack of fluency with a lack of mastery inevitably leads to biased diagnostic outcomes. For instance, a student may have fully understood fraction addition but still make a careless error. A traditional CDM would not only misdiagnose this as non-mastery but also trigger unnecessary interventions.

To overcome this limitation, as shown in the lower branch of Figure 1, we propose a paradigm shift from single-mastery diagnosis to joint modeling of mastery and fluency. Specifically, behavioral proxies (e.g., practice volume, historical accuracy, and response variability) are extracted from raw logs to build a fluency representation disentangled from cognitive mastery, enabling differentiated student state classification and precise, targeted interventions. To realize this paradigm, we propose MFCD, a dual-stream framework that extends exist-

ing CDMs by disentangling cognitive mastery and behavioral fluency. Our main contributions are summarized as follows:

- We propose MFCD, a novel and extensible CD framework that disentangles student profiles into mastery and fluency dimensions through multi-task learning.
- We formalize three behavioral proxy features, namely exposure intensity, empirical accuracy and response consistency, and design a dynamic fusion gate to adaptively integrate behavioral fluency with cognitive mastery.
- Extensive evaluations on six benchmark datasets with various CDM baselines clearly demonstrate the rationale and effectiveness of our MFCD.

2 Related Work

2.1 Psychometrics-based CDMs

Traditional psychometric-based CDMs can be categorized into ability-level and cognition-level paradigms. The ability-level paradigm focuses on macro-level holistic evaluation, modeling a student's overall ability as a continuous latent trait to predict response probability, without explicitly diagnosing specific knowledge components. Examples include Item Response Theory (IRT) [2] and its multidimensional extension (MIRT) [4], which characterize exercises via parameters such as discrimination, difficulty and guessing. In contrast, the cognition-level paradigm adopts a micro-level diagnostic perspective, modeling a student's knowledge state as a fine-grained binary attribute vector (i.e., mastered or not mastered) over specific knowledge concepts, as defined by the Q-matrix. For example, the Deterministic Input, Noisy "And" gate (DINA) model [5, 14] assumes that a correct response requires mastery of all concepts associated with the exercise.

Despite their theoretical interpretability, these traditional CDMs are often too rigid to capture complex and non-linear learning trajectories, and their parametric assumptions further constrain adaptability to high-dimensional interaction data.

2.2 Learning-based CDMs

In contrast to psychometrics-based models relying on strict theoretical assumptions, learning-based CDMs adopt a data-driven paradigm that directly learns latent cognitive states from historical interaction logs. These models are generally categorized into Machine Learning (ML)-based and Deep Learning (DL)-based approaches.

Early ML-based CDMs draw inspiration from collaborative filtering and matrix factorization, including matrix factorization [6] and its probabilistic variant [15], both of which factorize the student-exercise response matrix into low-dimensional latent trait vectors. Beyond matrix factorization, [7] applies support vector machine to classify mastery states via maximum-margin hyperplanes. [8]

employs a fuzzy set-theoretic framework that models continuous knowledge proficiency levels to accommodate both dichotomous and partially scored items. More recently, [9] enriches cognitive profiling by extracting an effort trait from interaction logs, and integrating it with ability traits and a knowledge weakness parameter within a compensation mechanism.

DL-based CDMs leverage deep neural networks to automatically capture complex, non-linear interactions between students and exercises. Neural CDM (NCDM) [10] introduces a neural framework with non-negative layers to enforce educational monotonicity, ensuring both accuracy and interpretability. To address sparse knowledge coverage, Knowledge-association based extension of the Neural Cognitive Diagnosis model (KaNCD) [11] models concept associations via low-dimensional embeddings, while Knowledge Concept Cold-Start Cognitive Diagnosis model (KCSCD) [16] targets concept cold-start via a gating-fused inference and ability module. Beyond accuracy improvement, CD research has also expanded to other critical issues, such as fairness. The Fair cognitive diagnosis WIthout Sensitive Attributes (FairWISA) [12] is a model-agnostic framework that mitigates unfairness without requiring sensitive attributes, via a max-min training strategy for pseudo-group inference and fairness regularization.

3 Problem Formulation

Let $\mathcal{U} = \{u_1, u_2, \ldots, u_N\}$ denote the set of N students, and $\mathcal{E} = \{e_1, e_2, \ldots, e_M\}$ denote the set of M exercises, $\mathcal{C} = \{c_1, c_2, \ldots, c_K\}$ denote the set of K knowledge concepts, and $\mathcal{R} = \{(u_i, e_j, r_{i,j}) \mid u_i \in \mathcal{U}, e_j \in \mathcal{E}\}$ denote the set of response records, where binary response $r_{i,j}$ equals 0 for an incorrect answer or 1 for a correct answer. The exercise-concept correlation matrix is encoded by a binary matrix named the Q-matrix, denoted as $\boldsymbol{Q} = [q_{j,k}]_{M \times K}$. Specifically, $q_{j,k} = 1$ if concept c_k is required to correctly answer item e_j, and $q_{j,k} = 0$ otherwise.

Definition 1. *(Cognitive Diagnosis) Given the response records $\mathcal{R}$ and the Q-matrix $\boldsymbol{Q}$, cognitive diagnosis aims to estimate students' latent cognitive states from their interaction histories. These states are typically represented by knowledge proficiency $\alpha_{i,k}$ of student $u_i \in \mathcal{U}$ on concept $c_k \in \mathcal{C}$, along with auxiliary item parameters such as difficulty and discrimination.*

Definition 2. *Difficulty and discrimination parameters characterize exercise properties, and guessing and slip parameters capture inconsistencies between observed responses and latent cognitive states. Difficulty quantifies the baseline success probability for an average-proficiency student. Discrimination measures how sensitively the success probability responds to changes in cognitive states. Guessing quantifies the probability of a correct response despite lacking the corresponding knowledge state. Slip measures the probability of an incorrect response despite possessing the corresponding knowledge state.*

For each exercise $e_j \in \mathcal{E}$, its difficulty, discrimination guessing and slip parameters are typically denoted as $\boldsymbol{h}_j^{\text{diff}}, \boldsymbol{h}_j^{\text{disc}} \in \mathbb{R}^K, h_j^{\text{gues}}, h_j^{\text{slip}} \in \mathbb{R}$ ($j \in \{1, 2, \cdots, M\}$), respectively.

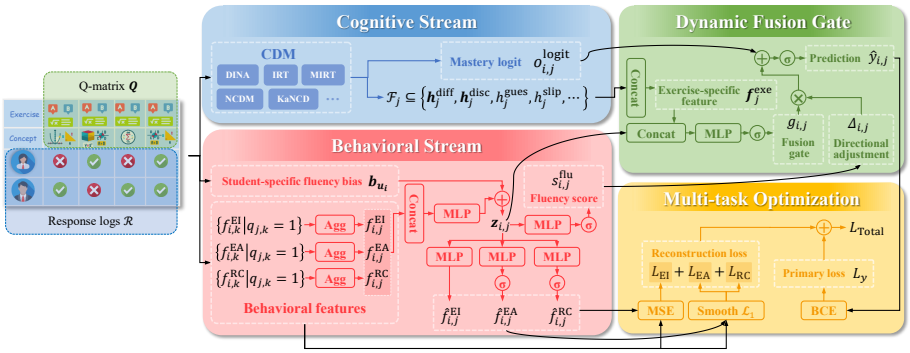


Fig. 2: The overall architecture of MFCD.

4 Methodology

Figure 2 illustrates the overall architecture of MFCD, which consists of mastery stream, behavioral stream, and dynamic fusion gate, jointly optimized under multi-task optimization to preserve behavioral information.

4.1 Cognitive Stream

According to the core monotonicity assumption of CD, we adopt a standard CDM (e.g., NCDM, DINA, or IRT) as the cognitive stream, which takes the response records $\mathcal{R}$ and Q-matrix $\boldsymbol{Q}$ as input. For a given student u_i and exercise e_j, the stream produces the mastery logit $o_{i,j}^{\text{logit}}$ of the base correct response probability as below:

$$\left\{o_{i,j}^{\text{logit}}, \mathcal{F}_j\right\} = \text{CDM}\left(u_i, e_j, \mathcal{R}, \boldsymbol{Q}\right) \tag{1}$$

where $\mathcal{F}_j \subseteq \left\{\boldsymbol{h}_j^{\text{diff}}, \boldsymbol{h}_j^{\text{disc}}, h_j^{\text{gues}}, h_j^{\text{slip}}, \cdots\right\}$ denotes the set of exercise-specific cognitive parameters provided by the standard CDM. For example, $\mathcal{F}_j = \left\{\boldsymbol{h}_j^{\text{diff}}, \boldsymbol{h}_j^{\text{disc}}\right\}$ for NCDM, and $\mathcal{F}_j = \left\{h_j^{\text{gues}}, h_j^{\text{slip}}\right\}$ for DINA.

4.2 Behavioral Stream

To capture procedural learning dynamics, we introduce three statistical behavioral features based on each student's historical interactions with each knowledge concept, namely exposure intensity, empirical accuracy and response consistency. Let $\mathcal{R}_{i,k} = \{(u_i, e_j, r_{i,j}) \mid (u_i, e_j, r_{i,j}) \in \mathcal{R},\ q_{j,k} = 1\}$ denote the subset of historical records for student u_i that are associated with concept c_k. These features are formalized using $\mathcal{R}_{i,k}$ as follows.

– **Exposure Intensity** quantifies the cumulative interaction density of a student with a specific concept.

$$f_{i,k}^{\mathrm{EI}} = \log\left(1 + |\mathcal{R}_{i,k}|\right) \tag{2}$$

where $|\mathcal{R}_{i,k}|$ is the size of the record subset $\mathcal{R}_{i,k}$.

– **Empirical Accuracy** reflects the observed historical correctness rate.

$$f_{i,k}^{\mathrm{EA}} = \frac{\sum_{(u_i,e_j,r_{i,j})\in\mathcal{R}_{i,k}} r_{i,j}}{|\mathcal{R}_{i,k}| + \epsilon} \tag{3}$$

where ϵ is a small smoothing constant (e.g., 10^{-5}) that prevents division by zero.

– **Response Consistency** measures the stability in the student's recent correctness pattern, capturing erratic behaviors indicative of slips or guesses.

$$f_{i,k}^{\mathrm{RC}} = 1 - 4 \cdot \mathrm{Var}\left(\{r_{i,j} \mid (u_i, e_j, r_{i,j}) \in \mathcal{R}_{i,k}\}\right) \tag{4}$$

where $\mathrm{Var}(\cdot)$ denotes the variance operator, measuring the dispersion of the correctness values.

Then, the above three behavioral features are concatenated and processed by a multi-layer perceptron (MLP) to obtain a latent fluency state $\boldsymbol{z}_{i,j} \in \mathbb{R}^D$:

$$\boldsymbol{z}_{i,j} = \mathrm{MLP}_{\mathrm{beh}}\left(\left[f_{i,j}^{\mathrm{EI}} \parallel f_{i,j}^{\mathrm{EA}} \parallel f_{i,j}^{\mathrm{RC}}\right]\right) + \tanh\left(\boldsymbol{b}_{u_i}\right) \tag{5}$$

where $\boldsymbol{b}_{u_i} \in \mathbb{R}^D$ is a learnable student-specific bias term that captures the intrinsic fluency tendency of student u_i, independent of the current exercise context. Here,

$$f_{i,j}^{\mathrm{EI}} = \mathrm{Agg}\left(\left\{f_{i,k}^{\mathrm{EI}} \;\middle|\; q_{j,k} = 1, k \in \{1,\ldots,K\}\right\}\right), \tag{6}$$

$$f_{i,j}^{\mathrm{EA}} = \mathrm{Agg}\left(\left\{f_{i,k}^{\mathrm{EA}} \;\middle|\; q_{j,k} = 1, k \in \{1,\ldots,K\}\right\}\right), \tag{7}$$

$$f_{i,j}^{\mathrm{RC}} = \mathrm{Agg}\left(\left\{f_{i,k}^{\mathrm{RC}} \;\middle|\; q_{j,k} = 1, k \in \{1,\ldots,K\}\right\}\right) \tag{8}$$

where $\parallel$ denotes vector concatenation, and $\mathrm{Agg}(\cdot)$ denotes an aggregation function (e.g., mean pooling, max pooling, or attention-based weighting). To obtain an interpretable continuous fluency score $s_{i,j}^{\mathrm{flu}} \in (0,1)$, we map the latent representation $\boldsymbol{z}_{i,j}$ through an MLP followed by a Sigmoid activation $\sigma(\cdot)$:

$$s_{i,j}^{\mathrm{flu}} = \sigma\left(\mathrm{MLP}_{\mathrm{flu}}\left(\boldsymbol{z}_{i,j}\right)\right) \tag{9}$$

In addition, to support the following multi-task regularization objective, we derive three auxiliary reconstruction heads from the same latent representation $\boldsymbol{z}_{i,j}$ as follows:

$$\hat{f}_{i,j}^{\mathrm{EI}} = \mathrm{MLP}_{\mathrm{EI}}\left(\boldsymbol{z}_{i,j}\right), \tag{10}$$

$$\hat{f}_{i,j}^{\mathrm{EA}} = \sigma\left(\mathrm{MLP}_{\mathrm{EA}}\left(\boldsymbol{z}_{i,j}\right)\right), \tag{11}$$

$$\hat{f}_{i,j}^{\mathrm{RC}} = \sigma\left(\mathrm{MLP}_{\mathrm{RC}}\left(\boldsymbol{z}_{i,j}\right)\right) \tag{12}$$

These auxiliary heads are then optimized via regression losses to reconstruct their corresponding target proxies, forcing $\boldsymbol{z}_{i,j}$ to retain comprehensive information across all three behavioral dimensions.

4.3 Dynamic Fusion Gate

To enable adaptive integration of behavioral and cognitive information, we introduce a dynamic fusion gate $g_{i,j} \in (0,1)$ for each student-exercise pair (u_i, e_j). This gate captures the non-linear interaction between the behavioral latent representation $\boldsymbol{z}_{i,j}$ and the exercise-specific cognitive feature $\boldsymbol{f}_j^{\text{exe}}$:

$$g_{i,j} = \sigma\left(\text{MLP}_{\text{gate}}\left(\left[\boldsymbol{z}_{i,j} \parallel \boldsymbol{f}_j^{\text{exe}}\right]\right)\right) \tag{13}$$

where

$$\boldsymbol{f}_j^{\text{exe}} = \mathop{\Big\|}_{f \in \mathcal{F}_j} f. \tag{14}$$

The final prediction $\hat{y}_{i,j}$ for student u_i on exercise e_j is then calibrated via a residual correction mechanism as below:

$$\hat{y}_{i,j} = \sigma\left(o_{i,j}^{\text{logit}} + \gamma \cdot g_{i,j} \cdot \Delta_{i,j}\right) \tag{15}$$

where $\Delta_{i,j} = \left(s_{i,j}^{\text{flu}} - 0.5\right) \times 4.0 \in (-2, 2)$ is the directional adjustment term, and γ is a trainable scaling parameter controlling the overall magnitude of correction. The gate $g_{i,j}$ controls the fusion degree between the two streams, from pure CD (i.e., $g_{i,j} \to 0$) to full behavioral intervention (i.e., $g_{i,j} \to 1$).

4.4 Multi-Task Optimization

To prevent the behavioral stream from collapsing under the primary loss, we adopt a multi-task optimization strategy that reconstructs the three proxy features from the latent representation $\boldsymbol{z}_{i,j}$. The total loss consists of a primary loss and three auxiliary reconstruction losses:

$$L_{\text{Total}} = (1-\lambda) \cdot L_y + \lambda \cdot (L_{\text{EI}} + L_{\text{EA}} + L_{\text{RC}}) \tag{16}$$

where λ balances the primary and auxiliary losses, and

$$L_y = -\mathbb{E}_{(i,j)}\left[r_{i,j} \log(\hat{y}_{i,j}) + (1 - r_{i,j}) \log(1 - \hat{y}_{i,j})\right], \tag{17}$$

$$L_{\text{EI}} = \mathbb{E}_{(i,j)}\left[\left(\hat{f}_{i,j}^{\text{EI}} - f_{i,j}^{\text{EI}}\right)^2\right], \tag{18}$$

$$L_{\text{EA}} = \mathbb{E}_{(i,j)}\left[\text{smooth}_{\mathcal{L}_1}\left(\hat{f}_{i,j}^{\text{EA}}, f_{i,j}^{\text{EA}}\right)\right], \tag{19}$$

$$L_{\text{RC}} = \mathbb{E}_{(i,j)}\left[\text{smooth}_{\mathcal{L}_1}\left(\hat{f}_{i,j}^{\text{RC}}, f_{i,j}^{\text{RC}}\right)\right] \tag{20}$$

where $\mathbb{E}[\cdot]$ denotes the batch-wise average over samples and $\text{smooth}_{\mathcal{L}_1}(\cdot)$ denotes the smooth $\mathcal{L}_1$ loss [17].

5 Experiments

In this section, we present comprehensive experimental results to validate the effectiveness of MFCD, including comparison experiments, ablation study, hyperparameter sensitivity, and stability analysis.

Table 1: Comprehensive statistics of the benchmark datasets.

Dataset	#Student	#Exercise	#Concept	#Interaction
ASSIST09	4,163	17,746	123	>300K
ASSIST17	1,709	3,162	102	>300K
Junyi	10,000	714	40	>218K
FrcSub	536	20	8	>10K
Math1	4,209	20	11	>14K
Math2	3,911	20	16	>19K

5.1 Experimental Setup

Datasets All experiments are conducted on six public benchmark KT datasets, i.e. ASSIST09[4], ASSIST17[5], FrcSub[6], Math1[3], Math2[3], and Junyi[7]. Table 1 summarizes statistics of each dataset, including the numbers of students, exercises, knowledge concepts, and interactions. These datasets present varying degrees of cognitive difficulty, spatial scale, and interaction sparsity, ensuring our evaluation is robust across completely different scenarios. Each dataset is split into training (70%), validation (10%) and test (20%) sets.

Baselines We choose three psychometrics-based models, including IRT[8] [2], MIRT[8] [4] and DINA[8] [5, 18], as well as four DL-based models, namely NCDM[8] [10], KaNCD[8] [11], FairWISA[9] [12] and KCSCD[10] [16], as both the baselines and mastery backbone for MFCD[11]. All baseline methods are implemented using their official configurations.

Evaluation Metrics All methods are evaluated via four commonly-used metrics in KT field, including Area Under the ROC Curve (AUC), Accuracy (ACC), F1 Score (F1), and Root Mean Square Error (RMSE).

Implementation Details MFCD and all baseline models are implemented in PyTorch and optimized with Adam at a learning rate of 0.002, applying a uniform dropout of 0.5 to all hidden layers. Training is conducted on NVIDIA RTX 4090 GPUs.

[4] https://sites.google.com/site/assistmentsdata/home/assistment-2009-2010-data
[5] https://sites.google.com/view/assistmentsdatamining/dataset
[6] http://staff.ustc.edu.cn/~qiliuql/data/math2015.rar
[7] https://base.ustc.edu.cn/data/tests/junyi/
[8] https://github.com/bigdata-ustc/EduCDM
[9] https://github.com/HeFei-X/FairWISA
[10] https://github.com/lihihuan1213/KCSCD
[11] https://github.com/leemoomoo/MFCD

Table 2: Performance comparisons (%). Δ denotes the performance change of MFCD over the baseline. **Red bold** and green indicate improvement and degradation, respectively. ↑/↓ indicates that higher/lower is better.

Model		ASSIST09				ASSIST17				Junyi			
		AUC↑	ACC↑	F1↑	RMSE↓	AUC↑	ACC↑	F1↑	RMSE↓	AUC↑	ACC↑	F1↑	RMSE↓
IRT	baseline	69.85	69.03	79.70	45.02	77.64	82.14	89.85	35.64	67.81	75.28	85.44	41.95
	MFCD	71.05	69.80	79.35	44.49	89.61	86.60	92.01	30.45	80.38	79.87	87.60	37.98
	Δ	**1.20**	**0.77**	-0.35	**-0.53**	**11.97**	**4.46**	**2.16**	**-5.19**	**12.57**	**4.59**	**2.16**	**-3.97**
MIRT	baseline	65.68	65.56	74.51	49.23	73.43	77.13	85.78	43.27	60.92	68.66	79.37	51.94
	MFCD	76.40	73.10	80.10	43.40	90.70	87.79	92.64	29.61	77.37	78.71	86.42	39.10
	Δ	**10.72**	**7.54**	**5.59**	**-5.83**	**17.27**	**10.66**	**6.86**	**-13.66**	**16.45**	**10.05**	**7.05**	**-12.84**
DINA	baseline	58.89	53.10	59.57	55.15	67.19	51.05	63.43	48.28	66.39	55.63	66.48	49.48
	MFCD	75.04	69.70	77.31	46.76	84.27	80.30	88.41	36.38	78.34	76.13	84.90	41.77
	Δ	**16.15**	**16.60**	**17.74**	**-8.39**	**17.08**	**29.25**	**24.98**	**-11.90**	**11.95**	**20.50**	**18.42**	**-7.71**
NCDM	baseline	75.22	71.75	78.65	43.77	87.64	84.98	91.02	31.83	80.77	79.76	87.11	37.76
	MFCD	76.93	73.67	81.46	42.30	89.49	86.80	92.25	30.39	82.29	80.57	87.95	37.37
	Δ	**1.71**	**1.92**	**2.81**	**-1.47**	**1.85**	**1.82**	**1.23**	**-1.44**	**1.52**	**0.81**	**0.84**	**-0.39**
KaNCD	baseline	76.80	73.61	81.23	42.21	88.86	85.96	91.45	31.13	81.28	80.16	87.53	37.38
	MFCD	77.08	73.66	81.92	42.18	89.11	86.67	92.11	30.45	81.97	80.05	87.80	37.35
	Δ	**0.28**	**0.05**	**0.69**	**-0.03**	**0.25**	**0.71**	**0.66**	**-0.68**	**0.69**	-0.11	**0.27**	**-0.03**
FairWISA	baseline	71.75	68.57	76.29	45.54	87.95	85.43	91.29	31.97	80.02	79.65	87.19	37.99
	MFCD	71.70	68.67	76.59	45.51	88.20	85.53	91.34	31.86	79.72	79.73	87.31	38.08
	Δ	-0.05	**0.10**	**0.30**	**-0.03**	**0.25**	**0.10**	**0.05**	**-0.11**	-0.30	**0.08**	**0.12**	0.09
KCSCD	baseline	76.10	71.95	81.59	43.38	89.28	86.50	92.03	30.55	81.84	80.57	88.01	37.60
	MFCD	76.51	71.64	81.60	42.44	90.03	86.99	92.30	30.15	82.00	80.67	87.96	37.28
	Δ	**0.41**	-0.31	**0.01**	**-0.94**	**0.75**	**0.49**	**0.27**	**-0.40**	**0.16**	**0.10**	-0.05	**-0.32**
		Math1				Math2				FrcSub			
		AUC↑	ACC↑	F1↑	RMSE↓	AUC↑	ACC↑	F1↑	RMSE↓	AUC↑	ACC↑	F1↑	RMSE↓
IRT	baseline	74.48	67.28	71.43	45.42	73.67	66.97	72.23	45.59	49.11	52.33	65.54	58.32
	MFCD	79.91	72.46	74.82	43.98	78.99	71.19	75.31	43.64	87.67	78.64	81.81	38.60
	Δ	**5.43**	**5.18**	**3.39**	**-1.44**	**5.32**	**4.22**	**3.08**	**-1.95**	**38.56**	**26.31**	**16.27**	**-19.72**
MIRT	baseline	54.51	53.27	53.01	64.02	70.70	65.33	67.91	48.91	73.50	66.70	68.82	53.76
	MFCD	80.75	72.88	72.56	42.63	78.17	71.10	72.82	44.08	86.92	80.13	81.36	38.28
	Δ	**26.24**	**19.61**	**19.55**	**-21.39**	**7.47**	**5.77**	**4.91**	**-4.83**	**13.42**	**13.43**	**12.54**	**-15.48**
DINA	baseline	75.23	53.54	21.45	51.85	68.69	48.30	22.31	53.84	58.82	54.85	36.32	56.85
	MFCD	77.31	57.40	33.96	50.20	72.48	59.88	54.11	51.64	86.18	75.56	73.43	42.73
	Δ	**2.08**	**3.86**	**12.51**	**-1.65**	**3.79**	**11.58**	**31.80**	**-2.20**	**27.36**	**20.71**	**37.11**	**-14.12**
NCDM	baseline	79.32	71.23	74.19	43.74	78.64	70.97	73.05	44.11	88.81	81.53	83.21	37.61
	MFCD	80.85	72.13	75.12	43.92	79.15	71.18	74.42	43.48	89.80	84.79	86.31	35.57
	Δ	**1.53**	**0.90**	**0.93**	0.18	**0.51**	**0.21**	**1.37**	**-0.63**	**0.99**	**3.26**	**3.10**	**-2.04**
KaNCD	baseline	79.74	71.47	73.72	42.74	79.20	71.26	73.33	43.10	89.86	83.40	84.28	35.27
	MFCD	81.25	72.89	73.43	42.05	79.34	71.64	74.31	42.94	90.29	83.02	85.30	34.78
	Δ	**1.51**	**1.42**	-0.29	**-0.69**	**0.14**	**0.38**	**0.98**	**-0.16**	**0.43**	-0.38	**1.02**	**-0.49**
FairWISA	baseline	79.99	71.45	69.23	43.44	78.02	70.43	71.95	44.19	86.39	78.17	78.01	39.10
	MFCD	81.16	72.56	71.67	42.05	79.10	71.81	74.60	43.37	87.63	79.85	80.85	37.91
	Δ	**1.17**	**1.11**	**2.44**	**-1.39**	**1.08**	**1.38**	**2.65**	**-0.82**	**1.24**	**1.68**	**2.84**	**-1.19**

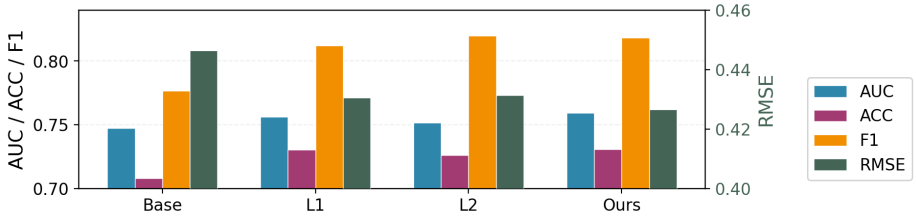


Fig. 3: Ablation study of MFCD.

5.2 Comparison Experiments

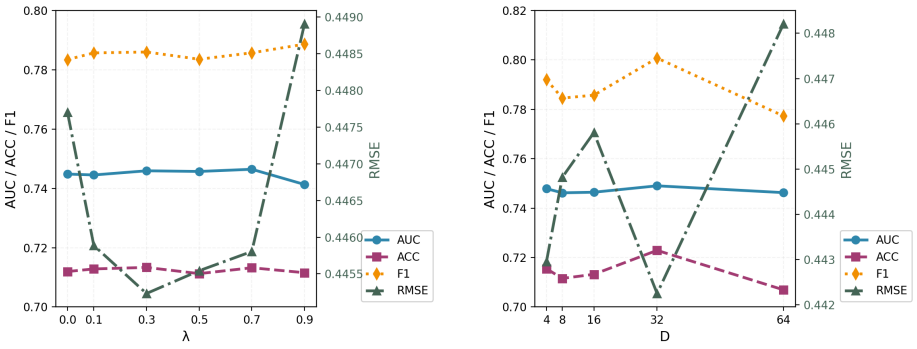
Table 2 reports the performance comparison on six benchmark datasets. As can be observed, MFCD consistently achieves superior or competitive performance against all baseline models across nearly all evaluation metrics and datasets. Notably, the performance gains are especially pronounced for traditional CDMs such as IRT, MIRT, and DINA, which lack inherent capacity for capturing complex behavioral dynamics. Even for state-of-the-art DL-based models such as KaNCD, FariWISA, and KCSCD, MFCD consistently yields performance improvements across most evaluation metrics, demonstrating its broad applicability and architectural compatibility. Overall, the comparison results consistently support the hypothesis that decoupling mastery from fluency provides a universally beneficial enhancement for CD.

5.3 Ablation Study

To evaluate each component of our method, we conduct an ablation study with four variants on ASSIST09 dataset. *Base* relies solely on the mastery backbone. *L1* augments *Base* with static fluency features and learns a static fusion of both mastery and fluency streams. *L2* further extends *L1* by incorporating multi-task learning to reconstruct the fluency proxy features. *Ours* (i.e., MFCD) additionally equips *L2* with the dynamic fusion gate. As shown in Figure 3, performance improves progressively from *Base* to *L1*, *L2*, and finally *Ours*, demonstrating the positive contribution of of fluency features, multi-task reconstruction, and dynamic fusion gate, respectively.

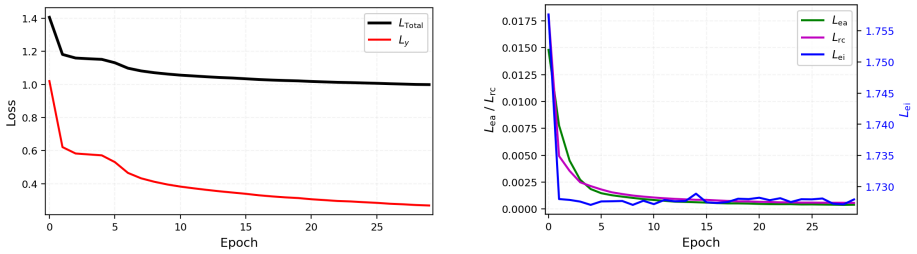
5.4 Hyperparameter Sensitivity Analysis

To investigate the impact of the weight λ and the fluency latent dimension size D on model performance, we conduct hyperparameter sensitivity analysis on the ASSIST09 dataset as shown in Figure 4. Figure 4a presents the AUC, ACC, F1 and RMSE on the ASSIST09 dataset for $\lambda \in \{0.0, 0.1, 0.3, 0.5, 0.7, 0.9\}$, where $\lambda = 0.0$ disables the multi-task reconstruction losses. As shown, MFCD achieves the best overall performance at $\lambda = 0.3$, achieving an optimal trade-off between the two learning objectives. Figure 4b reports the performance results



(a) (b)

Fig. 4: Sensitivity analysis of (a) weight λ and (b) dimension D of latent fluency state $\boldsymbol{z}_{i,k}$.



(a) (b)

Fig. 5: Loss curves of (a) L_{Total}, L_y, and (b) $L_{\text{ei}}, L_{\text{ea}}, L_{\text{rc}}$.

on the ASSIST09 dataset for $D \in \{4, 8, 16, 32, 64\}$. As shown, MFCD achieves best performance at $D = 32$, suggesting that this configuration strikes the most favorable balance between representational capacity and generalization.

5.5 Stability Analysis

To examine the training stability of MFCD, we monitor the loss curves and validation performance throughout the optimization process. As shown in Figure 5, all losses decrease steadily without conflict between objectives. The three auxiliary reconstruction losses converge rapidly, confirming that the multi-task setting effectively preserves behavioral information without compromising the primary prediction task. Meanwhile, the validation ACC in Figure 6 fluctuates slightly in later epochs, while the AUC remains relatively stable, indicating no severe overfitting and supporting early stopping near the peak. These results collectively verify the stability and effectiveness of jointly optimizing mastery and fluency.

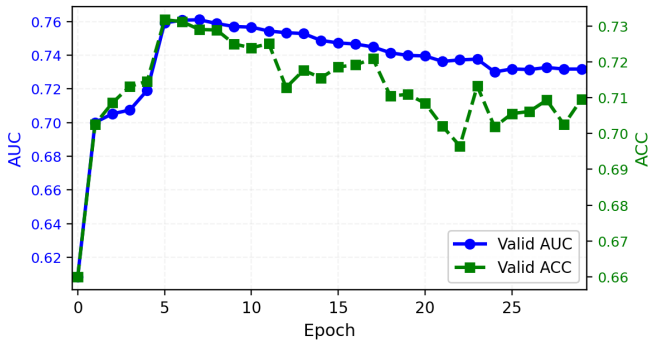


Fig. 6: Validation AUC and ACC curves.

5.6 Case Study

To validate the effectiveness of MFCD, we select six representative student-exercise pairs from the test set. Table 3 presents a detailed comparison between the backbone CDM and its corresponding MFCD variant. For each case, the table reports the set of knowledge concepts $\mathcal{C}_j = \{c_k \mid c_k \in \mathcal{C}, q_{j,k} = 1\}$ associated with exercise e_j, the three reconstructed behavioral proxy features $\hat{f}_{i,j}^{\mathrm{EI}}$, $\hat{f}_{i,j}^{\mathrm{EA}}$, $\hat{f}_{i,j}^{\mathrm{RC}}$, the learned fluency score $s_{i,j}^{\mathrm{flu}}$, the baseline prediction $y_{i,j}^{\mathrm{base}} = \mathrm{MLP}(o_{i,j}^{\mathrm{logit}})$ from the cognitive stream, the final prediction $\hat{y}_{i,j}$, and the ground-truth label $r_{i,j}$. The six cases are organized into three types based on the interaction pattern between the baseline cognitive prediction and the behavioral correction.

- In Cases 1, 2, and 4, the cognitive stream yields moderately high predictions despite clear behavioral signals of low fluency, such as insufficient practice (e.g., $\hat{f}_{i,j}^{\mathrm{EI}} = 1.386$ in Cases 2 and 4), extremely low accuracy (e.g., $\hat{f}_{i,j}^{\mathrm{EA}} = 0.056$ in Case 1), or insufficient observations for reliability (e.g., $\hat{f}_{i,j}^{\mathrm{RC}} = 0.000$ in Case 4). The behavioral stream applies substantial negative corrections accordingly, aligning predictions with the ground-truth incorrect responses. This demonstrates the behavioral stream's ability to prevent overestimation by distinguishing procedural deficiency from conceptual non-mastery.
- In Case 3, although the cognitive stream already yields a low prediction, the behavioral stream further lowers it by detecting a potential guessing pattern from limited practice, low accuracy, and high volatility. This confirms that behavioral evidence offers additional discriminative value even when the cognitive prediction is already pessimistic.
- In Cases 5 and 6, the behavioral stream identifies well-established procedural fluency from sufficient practice, high accuracy, and stable performance, and substantially elevates the conservative predictions of the cognitive stream to align with the ground-truth correct responses. This demonstrates the behavioral stream's ability to recover underestimation by identifying well-established procedural fluency that remains structurally invisible to the cognitive stream.

Table 3: Case studies of backbone CDM and the corresponding MFCD variant. Green and red indicate incorrect and correct predictions, respectively.

Case	u_i	e_j	$\mathcal{C}_j$	$\hat{f}^{\mathrm{EI}}_{i,j}$	$\hat{f}^{\mathrm{EA}}_{i,j}$	$\hat{f}^{\mathrm{RC}}_{i,j}$	$s^{\mathrm{flu}}_{i,j}$	$y^{\mathrm{base}}_{i,j}$ vs. $\hat{y}_{i,j}$	$r_{i,j}$
1	694	379	{2,25,32,49}	3.541	0.056	0.790	0.102	0.545 $\downarrow_{0.120}$	0
2	1244	6165	{33,34}	1.386	0.250	0.250	0.098	0.563 $\downarrow_{0.125}$	0
3	83580	65805	{50}	1.099	0.333	0.111	0.134	0.291 $\downarrow_{0.196}$	0
4	70729	58696	{79}	1.386	0.500	0.000	0.098	0.602 $\downarrow_{0.171}$	0
5	78587	89807	{307}	3.045	0.952	0.819	0.635	0.478 $\uparrow^{0.948}$	1
6	78566	89802	{307}	4.554	0.990	0.958	0.742	0.770 $\uparrow^{0.899}$	1

Overall, these cases suggest that MFCD can deliver more accurate, robust, and discriminative diagnosis via mastery-fluency disentanglement that separates conceptual deficits from procedural deficiencies.

6 Conclusion

In this paper, we proposed MFCD, a dual-stream CD framework that disentangles student cognitive states into mastery and behavioral fluency dimensions. Unlike existing CDMs that rely solely on binary correctness, MFCD captures behavioral fluency through three proxy features, including exposure intensity, empirical accuracy, and response consistency. A dynamic fusion gate adaptively integrates this behavioral signal with cognitive mastery from any backbone CDM, enabling fluency-aware correction. The multi-task objective jointly optimizes the primary prediction and three auxiliary reconstruction losses to preserve behavioral information. Extensive experiments on six benchmark datasets clearly demonstrate both the effectiveness and generalizability of our MFCD.

Disclosure of Interests. The authors have no competing interests to declare that are relevant to the content of this article.

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